F18CF – Linear Algebra

Tutorial Sheet 8: Eigenvalues and eigenspaces

Getting started.

1. We consider the following matrix:

$$A = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 6 & -11 & 6 \end{pmatrix}.$$

   (a) Is 2 an eigenvalue of A? If so determine the dimension and a basis of the corresponding eigenspace.

   (b) Is 4 an eigenvalue of A? If so determine the dimension and a basis of the corresponding eigenspace.

2. We consider the following matrix:

$$A = \begin{pmatrix} 2 & 0 & 0 & 1 \\ 0 & 2 & 1 & 0 \\ 0 & 1 & 2 & 0 \\ 1 & 0 & 0 & 2 \end{pmatrix}.$$

   (a) Is 1 an eigenvalue of A? If so determine the dimension of the corresponding eigenspace.

   (b) Is 3 an eigenvalue of A? If so determine the dimension of the corresponding eigenspace.

   (c) Is A diagonalisable?

Problems.

3. **A linearly independent family.** We consider the vector space $\mathbb{R}^{\mathbb{N}}$ of real sequences. We consider the following map, called the *shift*:

$$S : \mathbb{R}^{\mathbb{N}} \to \mathbb{R}^{\mathbb{N}}, \quad \begin{pmatrix} u_0 \\ u_1 \\ u_2 \\ u_3 \\ \vdots \end{pmatrix} \mapsto \begin{pmatrix} u_1 \\ u_2 \\ u_3 \\ u_4 \\ \vdots \end{pmatrix}.$$

   (a) Show that S is a linear map.

(b) Show that for every $\lambda > 0$, the following sequence $u^\lambda \in \mathbb{R}^{\mathbb{N}}$ is an eigenvector of S:

$$u^\lambda := \begin{pmatrix} 1 \\ \lambda \\ \lambda^2 \\ \lambda^3 \\ \vdots \end{pmatrix},$$

What is the corresponding eigenvalue?

(c) Deduce that for every $0 < \lambda_1 < \cdots < \lambda_k$, the family $u^{\lambda_1}, \ldots, u^{\lambda_k}$ is linearly independent.

4. **A diagonalisation problem.** Let A be the $n \times n$ matrix (with $n \geq 2$) all whose entries are 1:

$$A = \begin{pmatrix} 1 & \cdots & 1 \\ \vdots & \ddots & \vdots \\ 1 & \cdots & 1 \end{pmatrix}.$$

(a) What is the rank of A?

(b) Deduce that 0 is an eigenvalue of A and calculate the dimension of the associated eigenspace.

(c) Show that the vector $(1 \ldots 1)^T \in \mathbb{R}^n$ is an eigenvector of A. What is the corresponding eigenvalue?

(d) Deduce that A is diagonalisable.

5. **Unique eigenvalues.** Let A be a diagonalisable $n \times n$ matrix, and suppose that A has a unique eigenvalue $\lambda \in \mathbb{R}$.

(a) Show that $A = \lambda I_n$.

(b) Is the following matrix diagonalisable?

$$\begin{pmatrix} 1 & 2 & 3 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{pmatrix}.$$

6. **Nilpotent matrices.** We say that an $n \times n$ matrix A is **nilpotent** if there exists $k \geq 1$ such that $A^k = 0_n$.

(a) Let A be an $n \times n$ matrix and let $k \geq 1$. Show that if λ is an eigenvalue of A, then λ^k is an eigenvalue of A^k.

(b) Show that the following matrix is nilpotent:

$$A = \begin{pmatrix} 2 & 2 & -2 \\ 5 & 1 & -3 \\ 1 & 5 & -3 \end{pmatrix}.$$

(c) Show that 0 is the only eigenvalue of a nilpotent matrix.

(d) Show that the only diagonalisable nilpotent matrix is the zero matrix.